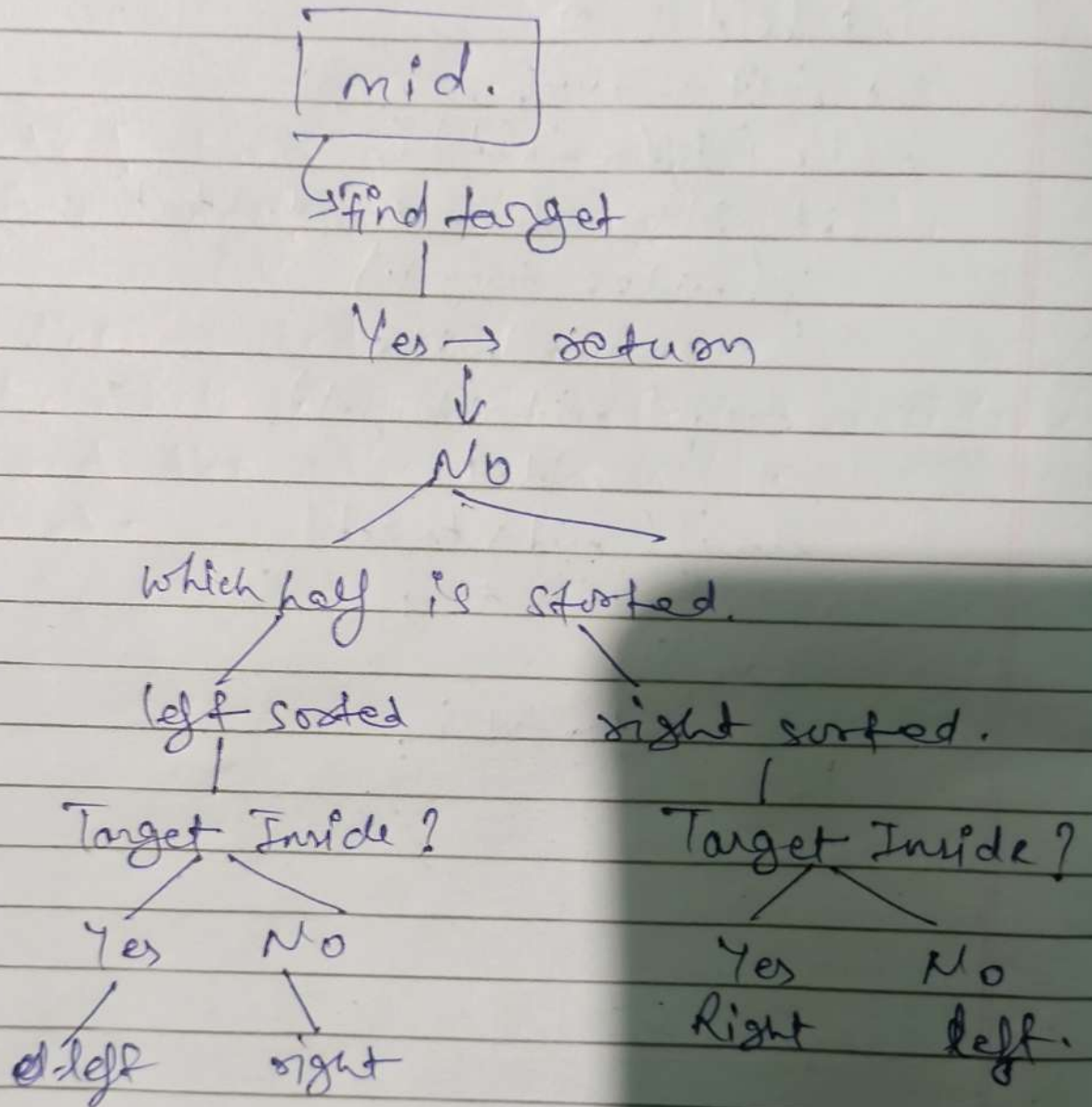


Q 10

nums = [4, 5, 6, 7, 0, 1, 2],
target = 0.



dry run

nums = [4, 5, 6, 7, 0, 1, 2]
target = 0

Let $\Rightarrow$ nums = [4, 5, 6, 7, 0, 1, 2]
start mid end

$$\text{mid} = \text{start} + (\text{end} - \text{start}) / 2$$
$$\text{mid} = 3$$

$\rightarrow$ kya left wala part sorted hai
 $\rightarrow$ Yes.

$\rightarrow$ ~~Tham~~ kya target 4 or 2 ke beech me hai. $\rightarrow$ No.

$\rightarrow$ No right :-

[0 1 2]

$\rightarrow$ target mil gaya.

Answer = 4.

* Time complexity = $O(\log n)$

* Space complexity = $O(1)$.

Trick :-

→ Normal Binary Search :-
↳ Whole array is sorted.

→ Rotated Binary Search :-
↳ One half is always sorted.

→ Find the sorted half.
↳ Check whether the target lies inside it.
↳ Discard the other half.
↳ Repeat.